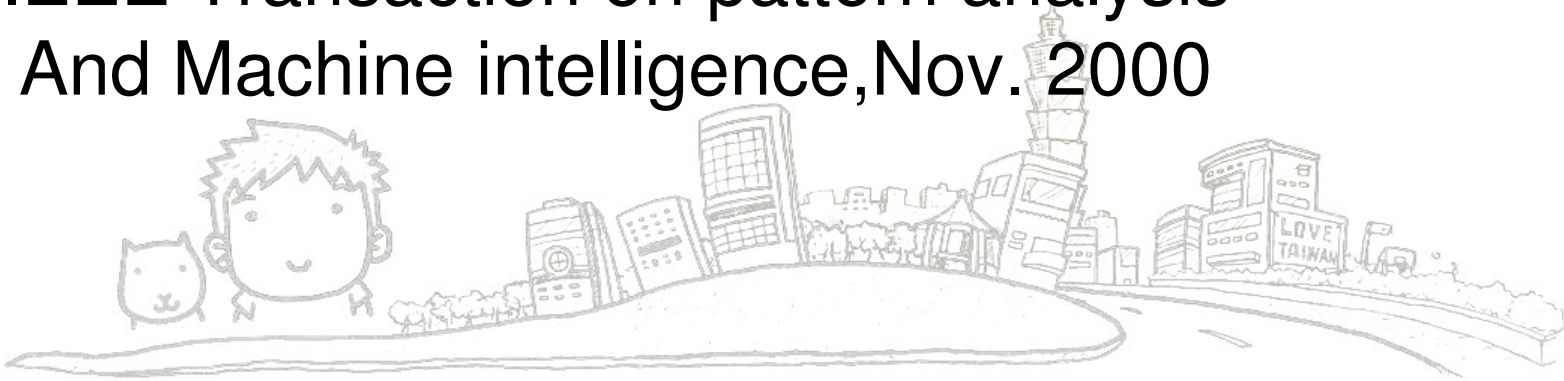
A Flexible New Technique for Camera Calibration

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Microsoft Research
IEEE Transaction on pattern analysis
And Machine intelligence,Nov. 2000



Outline

- Camera Calibration Technique
- Basic Equations
- Solving intrinsic parameters
- Solving extrinsic parameters
- Dealing with radial distortion
- Experimental Result
- Conclusion



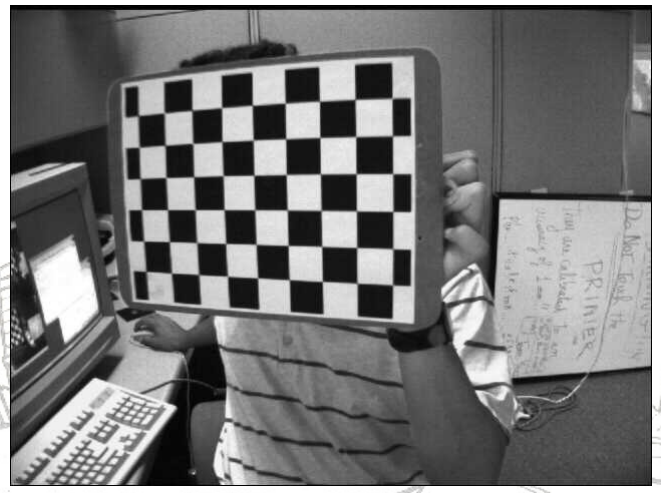
Camera Calibration Technique

- Photogrammetric calibration
  - Calibration Object consists of two or three planes orthogonal
- Self-Calibration
  - Moving a camera in a static scene



The Proposed Technique

- Camera to observe a planar pattern
  - At least two orientation
  - The planar pattern can move by hand
- The Technique
  - Between photogrammetric calibration
  - And self-calibration



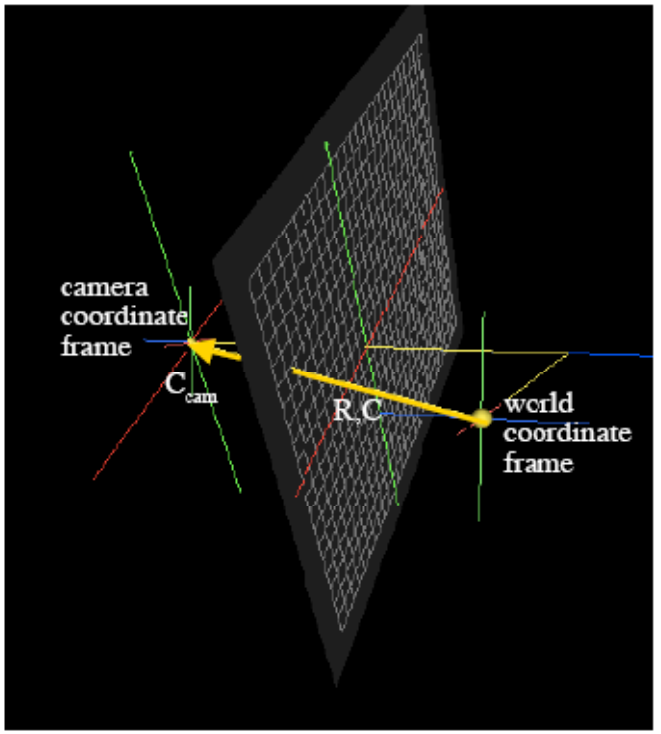


Basic Equations

$$s\tilde{\mathbf{m}} = \mathbf{A}\begin{bmatrix}\mathbf{R} & \mathbf{t}\end{bmatrix}\tilde{\mathbf{M}}$$

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = A[R \quad t]\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix}$$





Homography

- Z=0

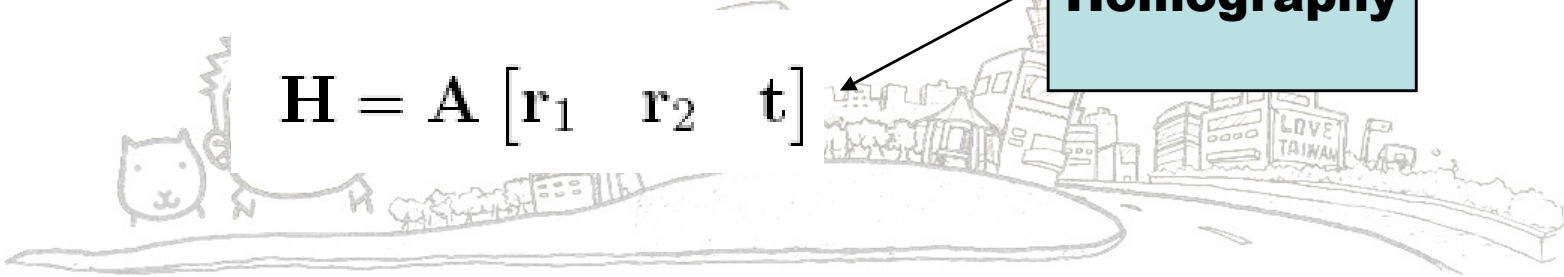
$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = A[R \quad t] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 & \mathbf{t} \end{bmatrix} \begin{bmatrix} X \\ Y \\ 0 \\ 1 \end{bmatrix}$$

$$= \mathbf{A} \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{t} \end{bmatrix} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} .$$

$$\mathbf{H} = \mathbf{A} \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{t} \end{bmatrix}$$

Homography



Estimated Homography

- Maximum Likelihood Criterion
  - Levenbergbe-Marquardt Algorithms
    - Steepest descent method
    - Newton Method

$$\sum_{i=1}^{n}\sum_{j=1}^{m} \|\mathbf{m}_{ij} - \hat{\mathbf{m}}(\mathbf{A}, \mathbf{R}_i, \mathbf{t}_i, \mathbf{M}_j)\|^2$$

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = A \begin{bmatrix} R & t \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

A,[R t] unknown

Solving intrinsic parameters

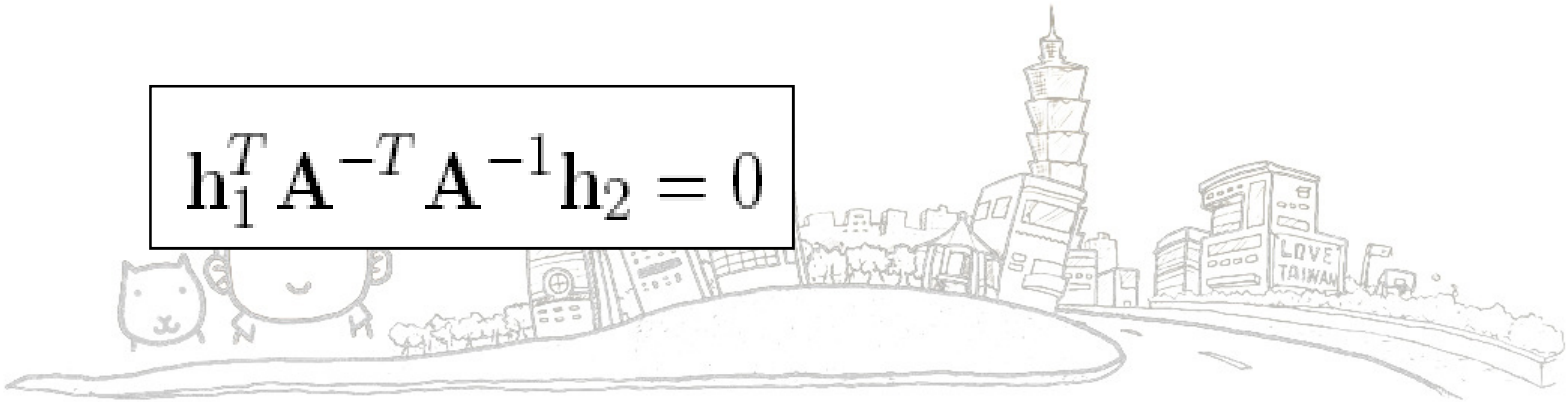
- Constraints intrinsic parameters

$$\mathbf{H} = \mathbf{A}\begin{bmatrix}\mathbf{r}_1 & \mathbf{r}_2 & \mathbf{t}\end{bmatrix}$$

$$\begin{bmatrix}\mathbf{h}_1 & \mathbf{h}_2 & \mathbf{h}_3\end{bmatrix} = \lambda\mathbf{A}\begin{bmatrix}\mathbf{r}_1 & \mathbf{r}_2 & \mathbf{t}\end{bmatrix}$$

r_1 r_2 are orthonormal

$$\mathbf{h}_1^T \mathbf{A}^{-T}\mathbf{A}^{-1}\mathbf{h}_2 = 0$$



Solving intrinsic parameters

- Singular Value Decomposition, SVD

$$\mathbf{H} = \mathbf{A}\begin{bmatrix}\mathbf{r}_1 & \mathbf{r}_2 & \mathbf{t}\end{bmatrix} \qquad \mathbf{A} = \begin{bmatrix}\alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1\end{bmatrix}$$

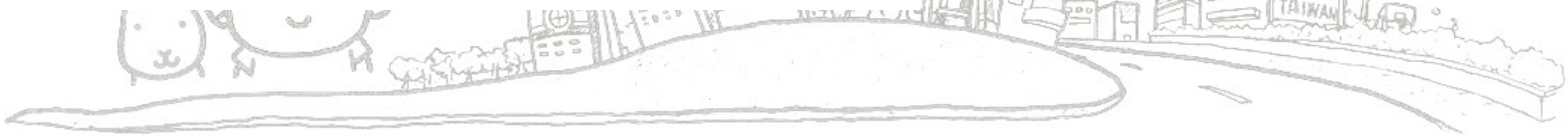
$$\mathbf{h}_1^T \mathbf{A}^{-T}\mathbf{A}^{-1}\mathbf{h}_2 = 0$$

$$\mathbf{h}_i^T \mathbf{B}\mathbf{h}_j = \mathbf{v}_{ij}^T \mathbf{b}$$

$$\mathbf{B} = \mathbf{A}^{-T}\mathbf{A}^{-1} \equiv \begin{bmatrix}B_{11} & B_{12} & B_{13} \\ B_{12} & B_{22} & B_{23} \\ B_{13} & B_{23} & B_{33}\end{bmatrix}$$

$$= \begin{bmatrix}\frac{1}{\alpha^2} & -\frac{\gamma}{\alpha^2\beta} & \frac{v_0\gamma - u_0\beta}{\alpha^2\beta} \\ -\frac{\gamma}{\alpha^2\beta} & \frac{\gamma^2}{\alpha^2\beta^2} + \frac{1}{\beta^2} & -\frac{\gamma(v_0\gamma - u_0\beta)}{\alpha^2\beta^2} - \frac{v_0}{\beta^2} \\ \frac{v_0\gamma - u_0\beta}{\alpha^2\beta} & -\frac{\gamma(v_0\gamma - u_0\beta)}{\alpha^2\beta^2} - \frac{v_0}{\beta^2} & \frac{(v_0\gamma - u_0\beta)^2}{\alpha^2\beta^2} + \frac{v_0^2}{\beta^2} + 1\end{bmatrix}$$

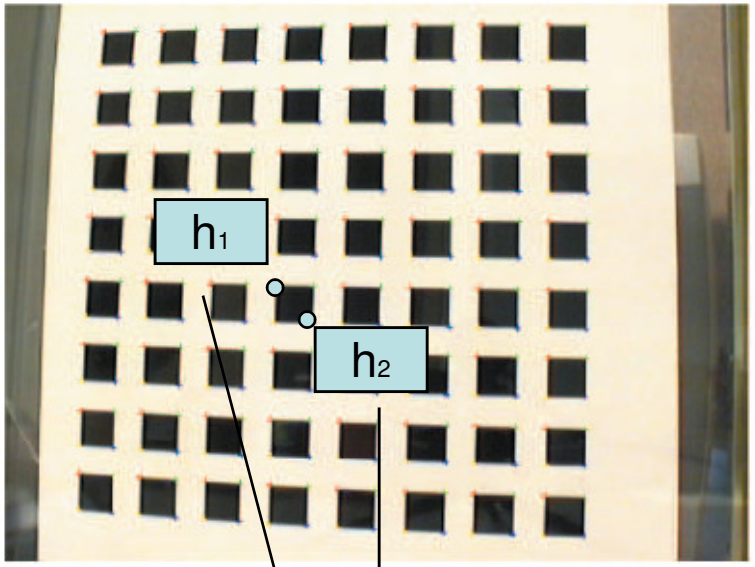
$$\mathbf{V}\mathbf{b} = 0$$



Solving intrinsic parameters

$$\mathbf{Vb} = 0$$

$$\begin{bmatrix} h_{11}h_{21} \\ h_{11}h_{22}+h_{12}h_{21} \\ h_{12}h_{22} \\ h_{13}h_{21}+h_{11}h_{23} \\ h_{13}h_{22}+h_{12}h_{23} \\ h_{13}h_{23} \end{bmatrix}^{\top} \begin{bmatrix} B_{11} \\ B_{12} \\ B_{22} \\ B_{13} \\ B_{23} \\ B_{33} \end{bmatrix} = O$$



$$[\mathbf{h}_1 \quad \mathbf{h}_2 \quad \mathbf{h}_3]$$

Solving intrinsic parameters

$$\mathbf{B} = \mathbf{A}^{-T}\mathbf{A}^{-1} \equiv \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{12} & B_{22} & B_{23} \\ B_{13} & B_{23} & B_{33} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\alpha^2} & -\frac{\gamma}{\alpha^2\beta} & \frac{v_0\gamma - u_0\beta}{\alpha^2\beta} \\ -\frac{\gamma}{\alpha^2\beta} & \frac{\gamma^2}{\alpha^2\beta^2} + \frac{1}{\beta^2} & -\frac{\gamma(v_0\gamma - u_0\beta)}{\alpha^2\beta^2} - \frac{v_0}{\beta^2} \\ \frac{v_0\gamma - u_0\beta}{\alpha^2\beta} & -\frac{\gamma(v_0\gamma - u_0\beta)}{\alpha^2\beta^2} - \frac{v_0}{\beta^2} & \frac{(v_0\gamma - u_0\beta)^2}{\alpha^2\beta^2} + \frac{v_0^2}{\beta^2} + 1 \end{bmatrix}$$

$$v_0 = (B_{12}B_{13} - B_{11}B_{23})/(B_{11}B_{22} - B_{12}^2)$$

$$\lambda = B_{33} - [B_{13}^2 + v_0(B_{12}B_{13} - B_{11}B_{23})]/B_{11}$$

$$\alpha = \sqrt{\lambda/B_{11}}$$

$$\beta = \sqrt{\lambda B_{11}/(B_{11}B_{22} - B_{12}^2)}$$

$$\gamma = -B_{12}\alpha^2\beta/\lambda$$

$$u_0 = \gamma v_0/\beta - B_{13}\alpha^2/\lambda \; .$$

$$\mathbf{A} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix}$$



Solving extrinsic parameters

$$\mathbf{H} = \mathbf{A}\begin{bmatrix}\mathbf{r}_1 & \mathbf{r}_2 & \mathbf{t}\end{bmatrix}$$

$$\begin{bmatrix}\mathbf{h}_1 & \mathbf{h}_2 & \mathbf{h}_3\end{bmatrix} = \lambda\mathbf{A}\begin{bmatrix}\mathbf{r}_1 & \mathbf{r}_2 & \mathbf{t}\end{bmatrix}$$

$$\mathbf{r}_1 = \lambda\mathbf{A}^{-1}\mathbf{h}_1$$
$$\mathbf{r}_2 = \lambda\mathbf{A}^{-1}\mathbf{h}_2$$
$$\mathbf{r}_3 = \mathbf{r}_1 \times \mathbf{r}_2$$
$$\mathbf{t} = \lambda\mathbf{A}^{-1}\mathbf{h}_3$$



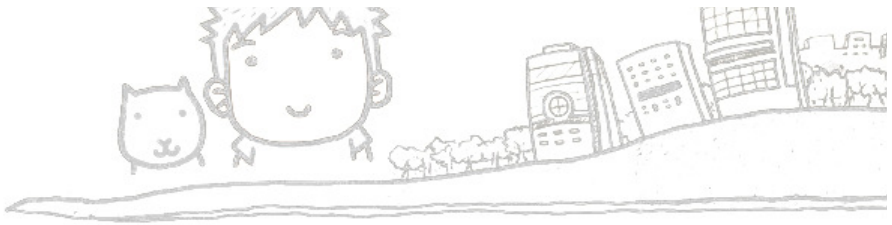
Dealing with radial distortion

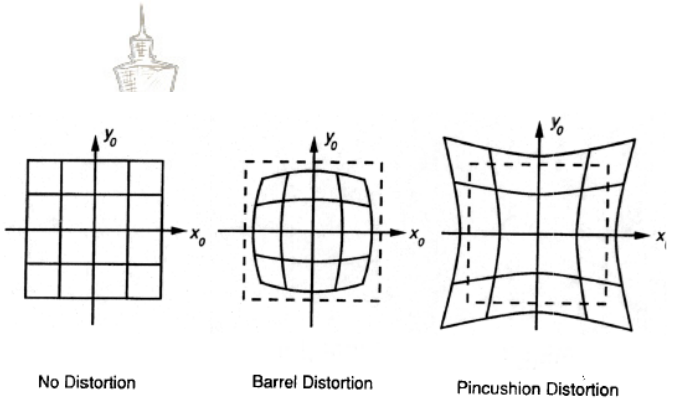
- x,y : ideal normalized image coordinates
- x,y bar : real normalized image coordinates

$$\breve{x} = x + x[k_1(x^2 + y^2) + k_2(x^2 + y^2)^2]$$
$$\breve{y} = y + y[k_1(x^2 + y^2) + k_2(x^2 + y^2)^2]$$

- u,v : ideal pixel image coordinates
- u,v bar : real observe image coordinates

$$\breve{u} = u + (u - u_0)[k_1(x^2 + y^2) + k_2(x^2 + y^2)^2]$$
$$\breve{v} = v + (v - v_0)[k_1(x^2 + y^2) + k_2(x^2 + y^2)^2]$$



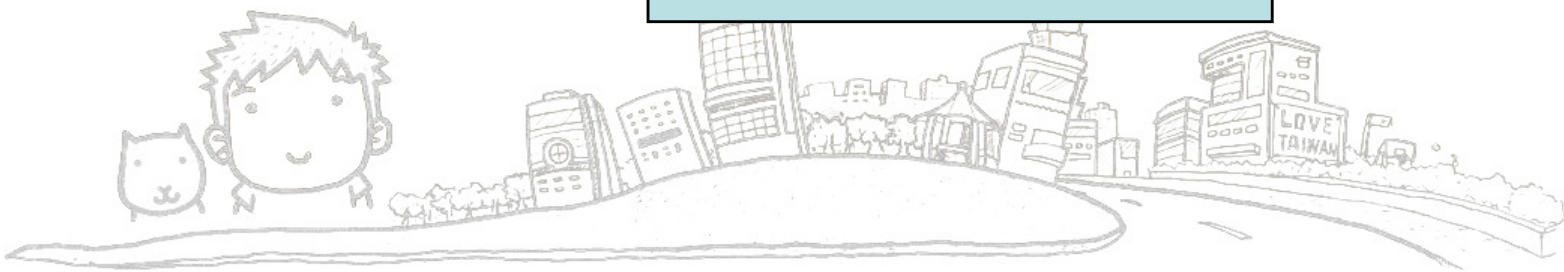


Dealing with radial distortion

$$\begin{bmatrix} (u-u_0)(x^2+y^2) & (u-u_0)(x^2+y^2)^2 \\ (v-v_0)(x^2+y^2) & (v-v_0)(x^2+y^2)^2 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} \breve{u}-u \\ \breve{v}-v \end{bmatrix}$$

$$\mathbf{k} = (\mathbf{D}^T\mathbf{D})^{-1}\mathbf{D}^T\mathbf{d}$$

Linear least-squares solution



Complete Maximum Likelihood Estimation

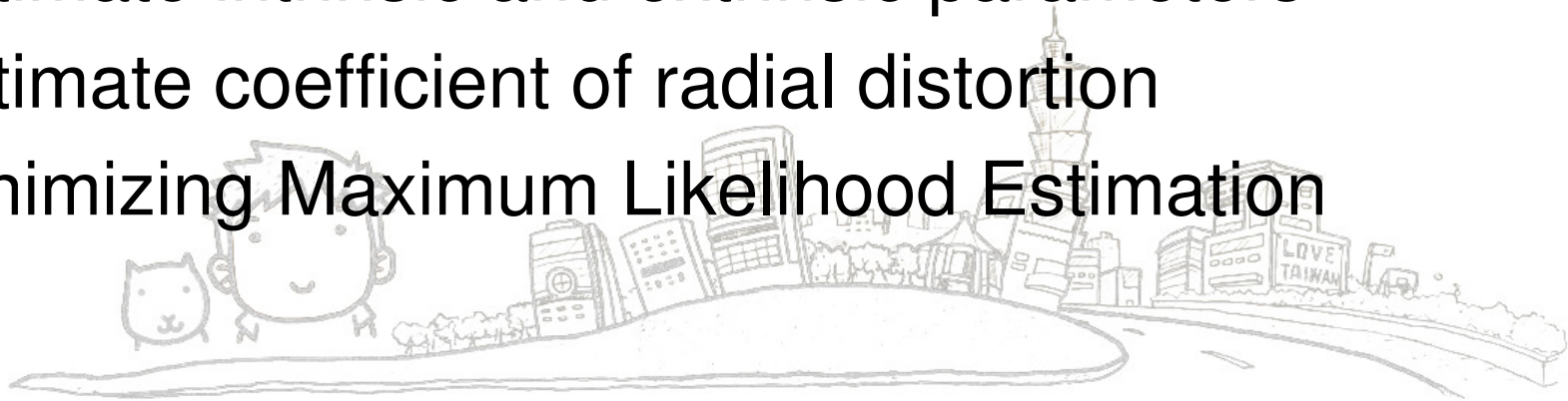
$$\sum_{i=1}^{n}\sum_{j=1}^{m} \|\mathbf{m}_{ij} - \breve{\mathbf{m}}(\mathbf{A}, k_1, k_2, \mathbf{R}_i, \mathbf{t}_i, \mathtt{M}_j)\|^2$$



Summary

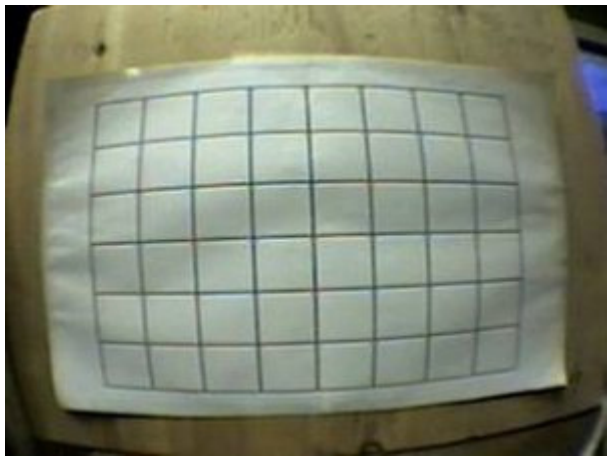
- Recommended calibration procedure

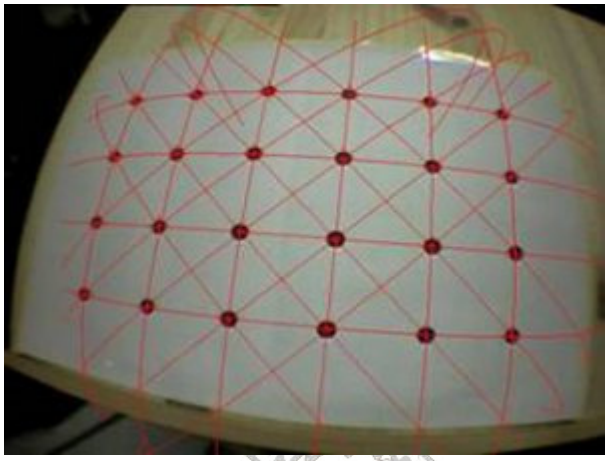
1) Print a pattern
   - attach it to a planar surface
2) Under different orientation
   - by moving plane or camera
3) Detect feature point
4) Estimate intrinsic and extrinsic parameters
5) Estimate coefficient of radial distortion
6) Minimizing Maximum Likelihood Estimation

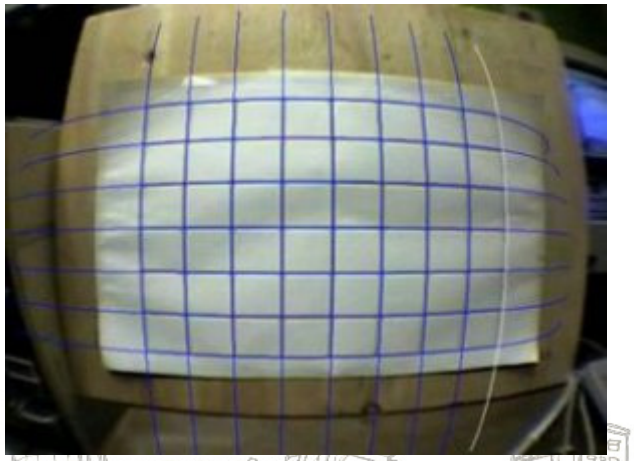


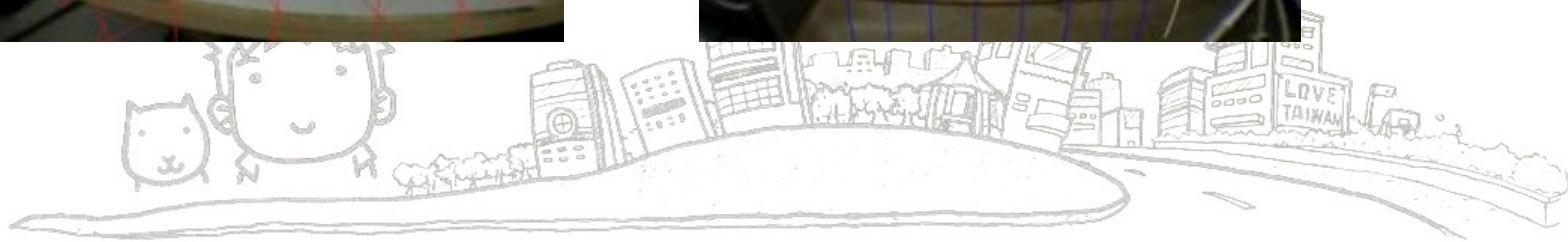
Camera Calibration











Experimental Result

- 8x8 squares,256 corners
- Resolution 640x480
- Pattern size 17cm x 17cm

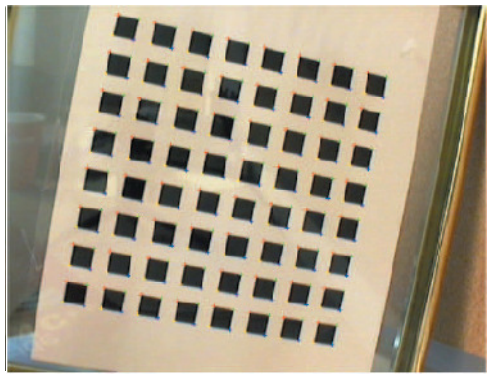
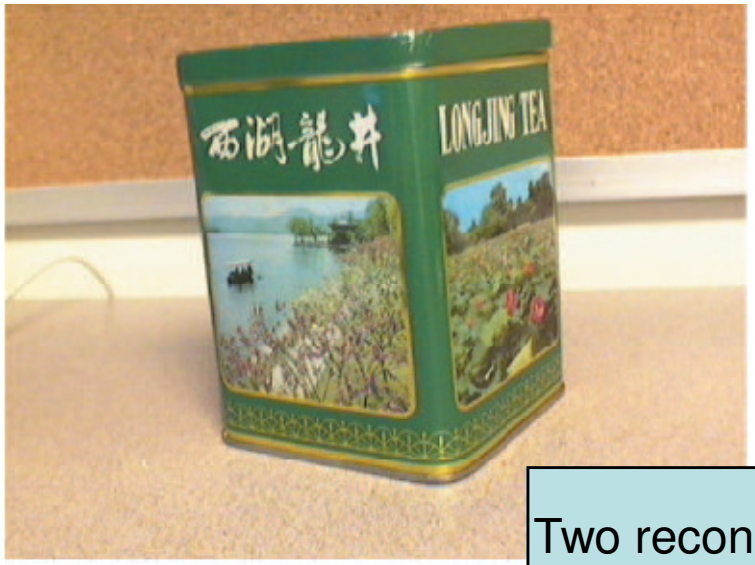


Table 1: Results with real data of 2 through 5 images

nb	2 images			3 images			4 images			5 images		
	initial	final	σ	initial	final	σ	initial	final	σ	initial	final	σ
α	825.59	830.47	4.74	917.65	830.80	2.06	876.62	831.81	1.56	877.16	832.50	1.41
β	825.26	830.24	4.85	920.53	830.69	2.10	876.22	831.82	1.55	876.80	832.53	1.38
γ	0	0	0	2.2956	0.1676	0.109	0.0658	0.2867	0.095	0.1752	0.2045	0.078
u_0	295.79	307.03	1.37	277.09	305.77	1.45	301.31	304.53	0.86	301.04	303.96	0.71
v_0	217.69	206.55	0.93	223.36	206.42	1.00	220.06	206.79	0.78	220.41	206.59	0.66
k_1	0.161	−0.227	0.006	0.128	−0.229	0.006	0.145	−0.229	0.005	0.136	−0.228	0.003
k_2	−1.955	0.194	0.032	−1.986	0.196	0.034	−2.089	0.195	0.028	−2.042	0.190	0.025
RMS	0.761	0.295		0.987	0.393		0.927	0.361		0.881	0.335	

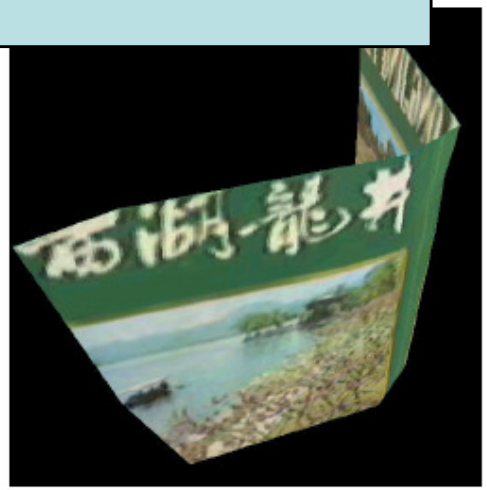
Application to image-base modeling

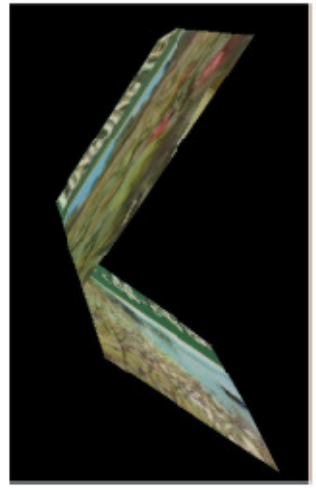




Two reconstruced planes
94.7°







Conclusion

- A few different orientations
- We can move either camera or planar pattern
- Motion doesn’t need to be known
- Consists of a closed-form solution
- Based on maximum likelihood criterion

